Embedding networks with the random walk first return time distribution

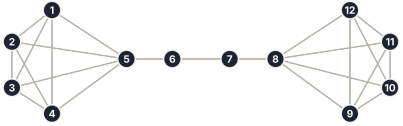
Vedanta Thapar, Renaud Lambiotte, and George T. Cantwell
University of Oxford · University of Cambridge

November 6, 2026

UNIVERSITY OF OXFORD
Mathematical Institute

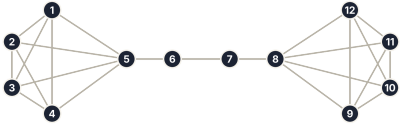
Oxford Mathematics

How should we represent a graph?

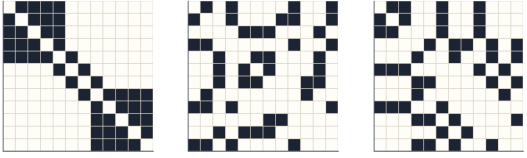


$$G = (V, E), \quad E \subseteq V \times V, \quad |V| = N, \ |E| = M$$

How should we represent a graph?

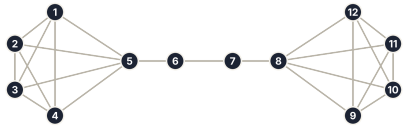


$$G = (V, E), \quad E \subseteq V \times V, \quad |V| = N, \ |E| = M$$

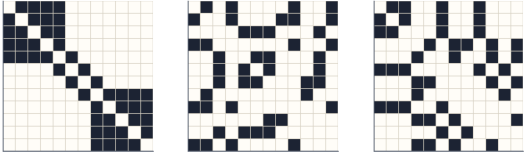


Adjacency matrix

How should we represent a graph?



$G = (V, E), \quad E \subseteq V \times V, \quad |V| = N, \ |E| = M$

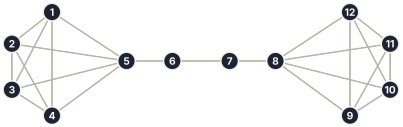


Adjacency matrix

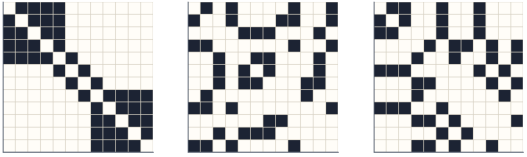
Embedding

$$\Phi : G \mapsto \Phi(G) \in \mathbb{R}^{N \times k}$$

How should we represent a graph?



$G = (V, E), \quad E \subseteq V \times V, \quad |V| = N, \; |E| = M$



Adjacency matrix

Embedding

$$\Phi : G \mapsto \Phi(G) \in \mathbb{R}^{N \times k}$$

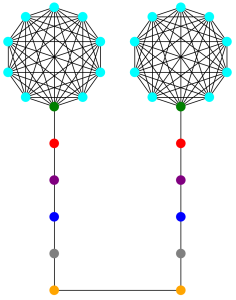
Ideal properties

- Simple to calculate and interpretable
- Invariant to node relabelling, i.e. Φ is permutation equivariant:

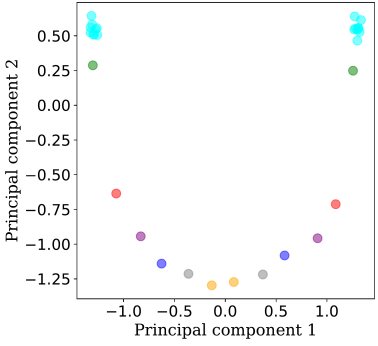
$$\Phi(PAP^\top) = P\,\Phi(A) \quad \forall P$$

- Invertible to a "unique" graph
- No hyperparameters
- Comparable across graphs: a natural distance $d\big(\Phi(G), \Phi(G')\big)$

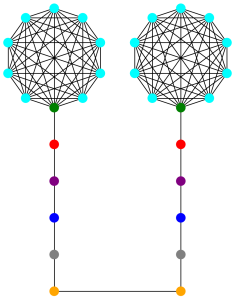
Proximity vs. structural embeddings

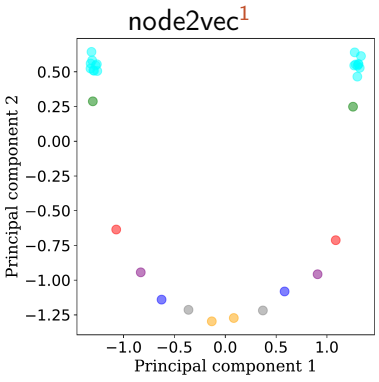


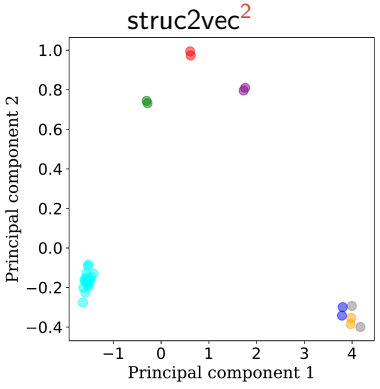
node2vec[1]



Proximity vs. structural embeddings



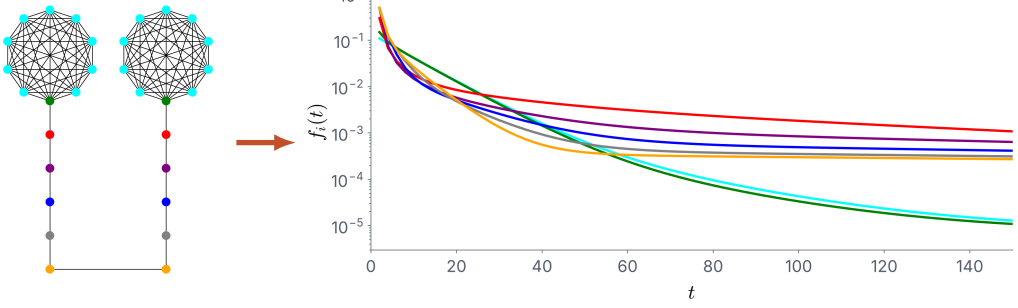




1 A. Grover and J. Leskovec, *node2vec: Scalable feature learning for networks*, KDD 2016, pp. 855–864.
2 L. F. R. Ribeiro, P. H. P. Saverese and D. R. Figueiredo, *struc2vec: Learning node representations from structural identity*, KDD 2017, pp. 385–394.

FRTD as a node embedding

$$f_i(t) = \mathbb{P}\big(u_t = i,\ u_s \neq i \text{ for } 0 < s < t \,\big|\, u_0 = i\big)$$



Early terms

- Mean (Kac's lemma): $\langle t \rangle_i = \dfrac{1}{\pi_i} = \dfrac{2m}{d_i}$
- $t = 2$: $f_i(2) = \dfrac{1}{d_i} \sum_{j \sim i} \dfrac{1}{d_j}$
- $t = 3$: $f_i(3) = \sum_{j,k} \dfrac{A_{ij} A_{jk} A_{ki}}{d_i d_j d_k}$

Theoretical analysis

The generating function

Mathematical Institute

Normalized adjacency $X = D^{-1/2} A D^{-1/2}.$

Spectral weight of node i $\mathsf{w}_i(\lambda) = \sum_{\alpha=1}^{N} \sum_{\beta=1}^{N_\alpha} (\psi_{\alpha\beta i})^2 \delta_{\lambda,\lambda_\alpha}$

First-return generating function $F_i(z) = \sum_{t \geq 1} z^t f_i(t)$

The generating function

Normalized adjacency $X = D^{-1/2} A D^{-1/2}.$

Spectral weight of node i $\mathrm{w}_i(\lambda) = \sum_{\alpha=1}^{N} \sum_{\beta=1}^{N_\alpha} (\psi_{\alpha\beta i})^2 \delta_{\lambda,\lambda_\alpha}$

First-return generating function $F_i(z) = \sum_{t \geq 1} z^t f_i(t)$

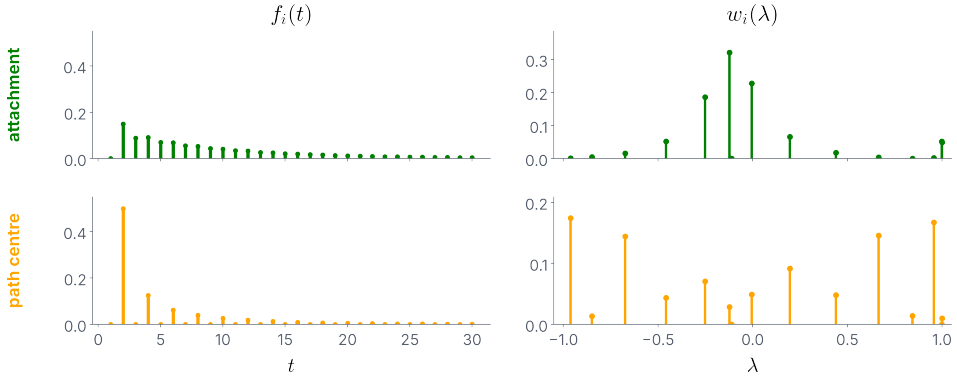
$$F_i(z) = 1 + \left(\sum_{\alpha} \frac{w_i(\lambda_\alpha)}{\lambda_\alpha z - 1} \right)^{-1}$$

FRTD equivalent nodes

Mathematical Institute

Node equivalence

$$f_i^{(G_1)} = f_j^{(G_2)} \iff w_i^{(G_1)}(\lambda) = w_j^{(G_2)}(\lambda) \quad \forall \lambda$$



FRTD equivalence of graphs

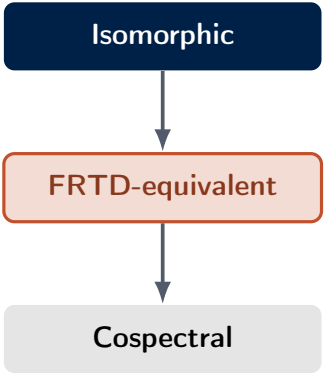
Isomorphic $\exists\, P:\ PA_1P^\top = A_2$

FRTD equivalence of graphs

Isomorphic $\exists\, P:\ PA_1P^\top = A_2$

Cospectral $\operatorname{spec}(X_1) = \operatorname{spec}(X_2)$

FRTD equivalence of graphs

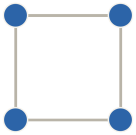


$\exists\, P:\ PA_1P^\top = A_2$

$\exists$ bijection $\pi:\ f_i^{(G_1)} = f_{\pi(i)}^{(G_2)}\ \ \forall i$

$\mathrm{spec}(X_1) = \mathrm{spec}(X_2)$

Same spectrum, different first returns

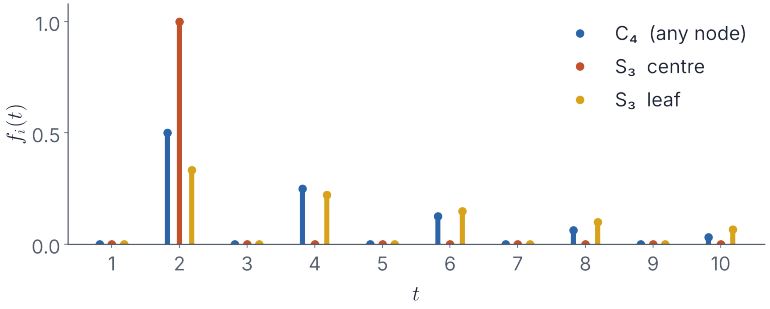


C_4

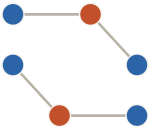


S_3

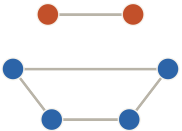
$\mathrm{spec}(X) = \{-1, 0, 0, 1\}$ for both



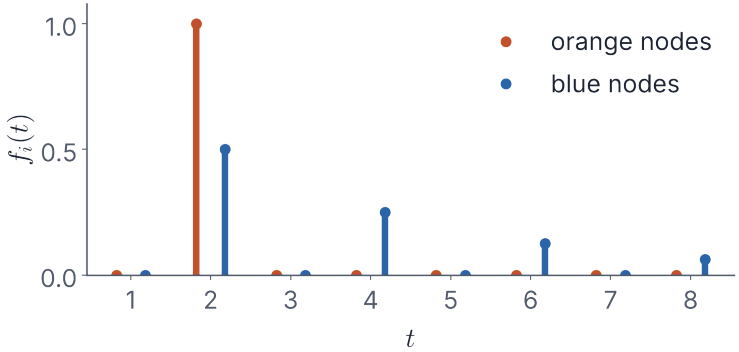
Different graphs, identical first returns



$2P_3$



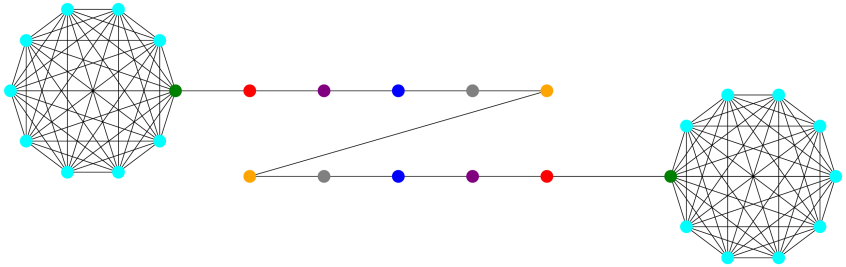
$K_2 \cup C_4$



Automorphisms

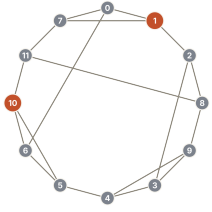
Mathematical Institute

i, j automorphically equivalent $\iff \exists$ permutation Π: $\Pi X = X\Pi,\ \Pi e_i = e_j$



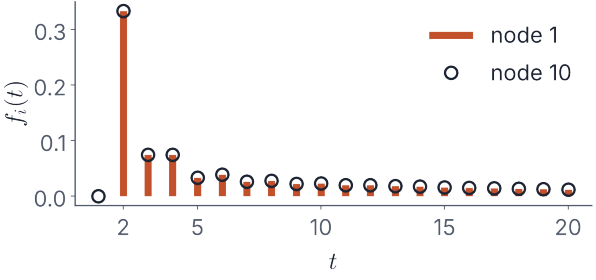
Automorphisms

i, j automorphically equivalent $\iff$ $\exists$ permutation Π: $\Pi X = X\Pi$, $\Pi e_i = e_j$



Frucht graph[1]

Nodes 1 and 10 have identical FRTDs:



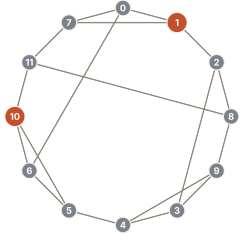
1 R. Frucht, *Graphs of degree three with a given abstract group*, Canadian Journal of Mathematics 1 (1949), pp. 365–378.

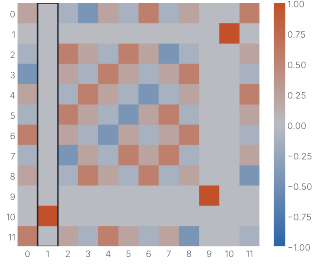
Automorphisms

Mathematical Institute

i, j automorphically equivalent $\iff \exists$ permutation Π: $\Pi X = X\Pi,\ \Pi e_i = e_j$

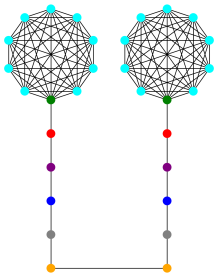
i, j FRTD-equivalent $\iff \exists$ orthogonal Q: $QX = XQ,\ Qe_i = e_j$





1 R. Frucht, *Graphs of degree three with a given abstract group*, Canadian Journal of Mathematics 1 (1949), pp. 365–378.

Defining distances



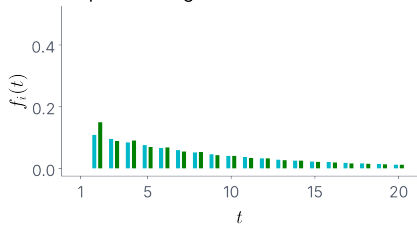
Nodes · total variation

$$d(f_i, f_j) = \tfrac{1}{2} \sum_t \left| f_i(t) - f_j(t) \right|$$

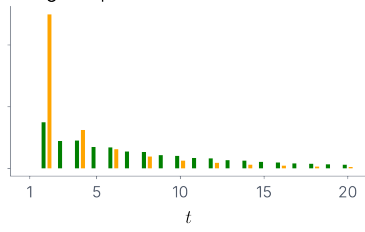
Graphs · best matching (Hungarian)

$$d(G_1, G_2) = \min_\pi \frac{1}{N} \sum_i d\left(f_i^{(G_1)}, f_{\pi(i)}^{(G_2)}\right)$$

clique vs bridge: d = 0.05



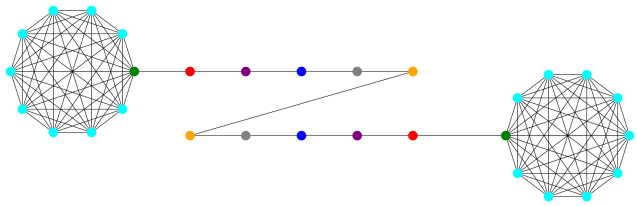
bridge vs path centre: d = 0.50



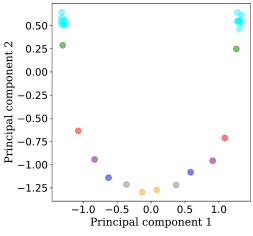
Empirical study

Role extraction

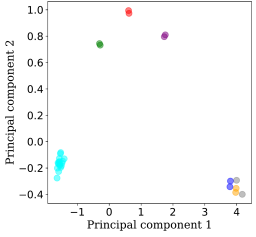
Mathematical Institute



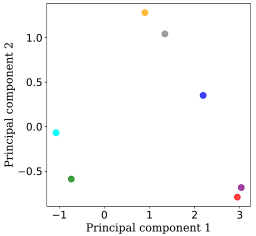
node2vec



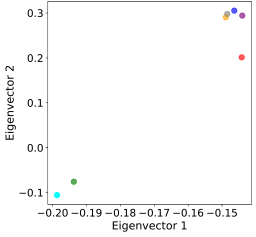
struc2vec



GraphWave



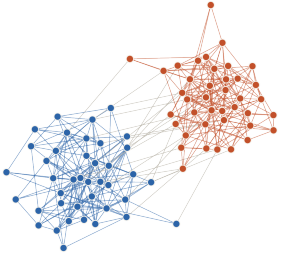
FRTD



Synthetic benchmarks

Mathematical Institute

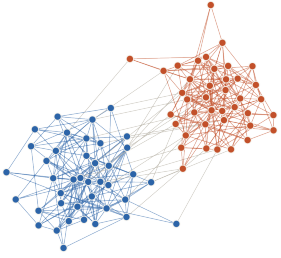
SBM · communities



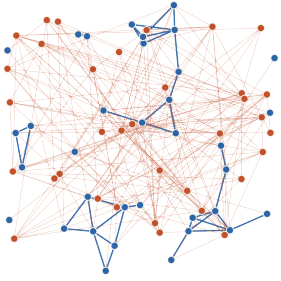
Synthetic benchmarks

University of Oxford
Mathematical Institute

SBM · communities



RG + **ER** · roles

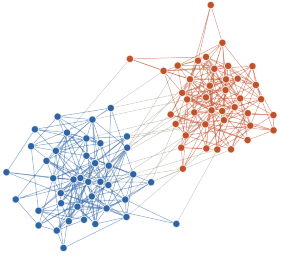


Blue nodes link only to nodes nearby in space; orange nodes link to anyone at random.

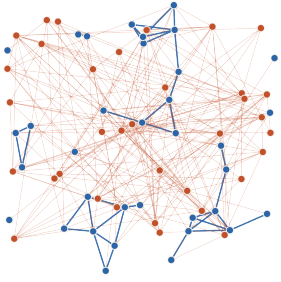
Synthetic benchmarks

University of Oxford
Mathematical Institute

SBM · communities

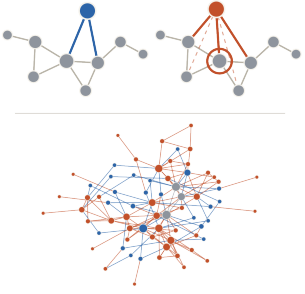


RG + **ER** · roles



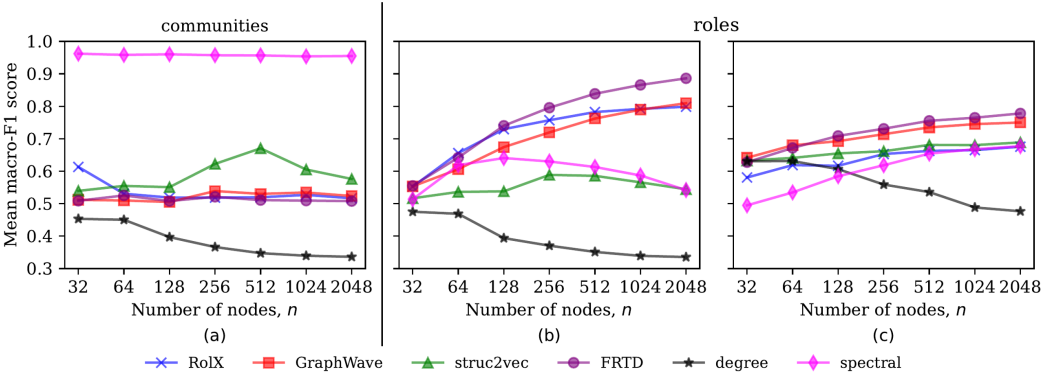
Blue nodes link only to nodes nearby in space; orange nodes link to anyone at random.

PA + **DD** · roles



A new node either joins high-degree nodes (PA) or copies a node and some of its edges (DD).

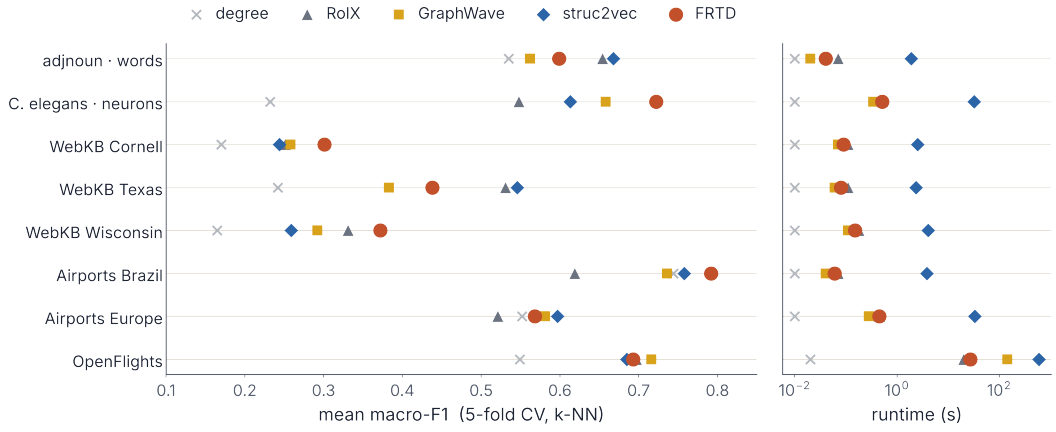
Results



Results from 5-fold stratified cross-validation with a k-nearest-neighbour classifier on each embedding.

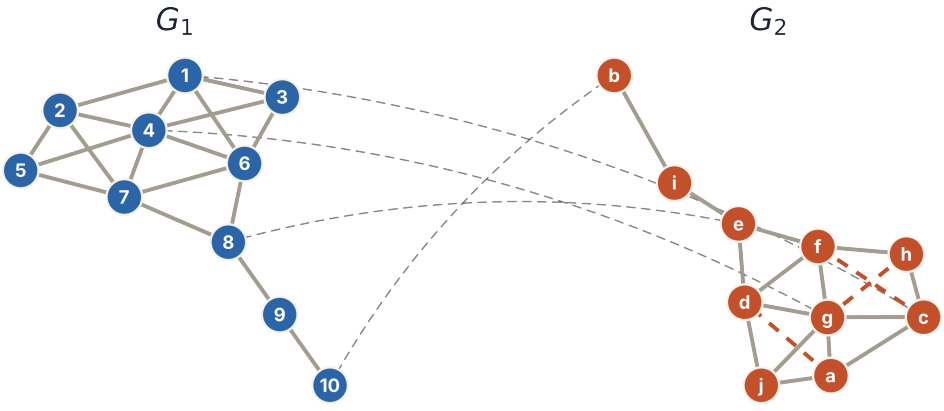
Real networks

Mathematical Institute



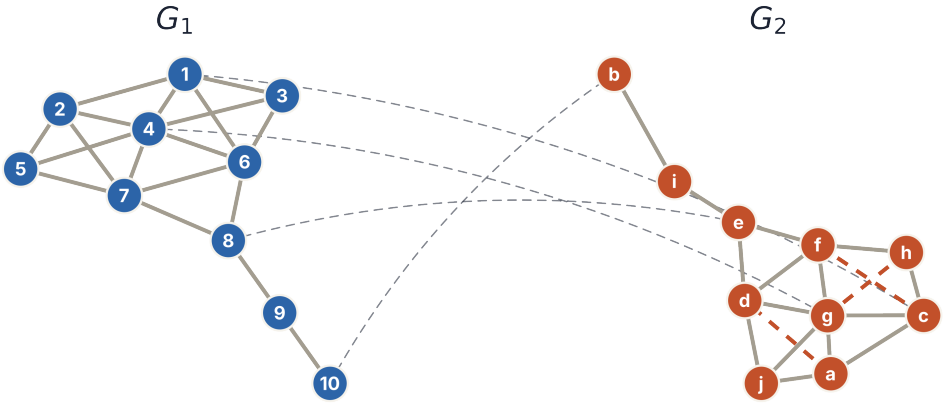
Metadata used as a proxy for roles: word type, neuron type, web-page type, airport size class.

Graph alignment



Graph alignment

Mathematical Institute



$$\pi^\star = \arg\min_{\pi} \left\| A_1 \Pi - \Pi A_2 \right\|_F^2$$

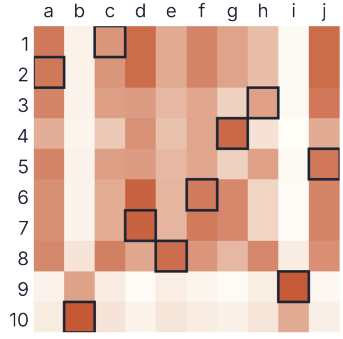
Practical solution

Linear assignment

$$\pi^\star = \arg\min_\pi \sum_i C\big(i, \pi(i)\big) = \arg\min_\Pi \operatorname{tr}\left(\Pi^\top C\right)$$

$C(i,j)$ = cost of matching node $i \in G_1$ to node $j \in G_2$.

Solvable exactly in $O(n^3)$ by the Hungarian algorithm.



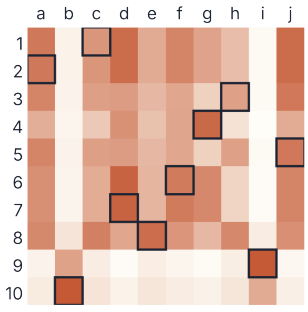
FRTD cost C for the example graphs (darker = cheaper). Boxes: true matches.

Practical solution

$$\min_{\pi} \left\| A_1 \Pi - \Pi A_2 \right\|_F^2 + \mu \operatorname{tr}\left(\Pi^\top C\right)$$

$C(i,j)$ = cost of matching node $i \in G_1$ to node $j \in G_2$.

- **FUGAL**[1]: $C(i,j)$ = squared Euclidean distance between feature vectors: degree, clustering coefficient, mean neighbour degree, mean neighbour clustering.
- **FUGAL-FRT**: $C(i,j) = d_{\mathrm{TV}}\left(f_i^{(G_1)}, f_j^{(G_2)}\right)$



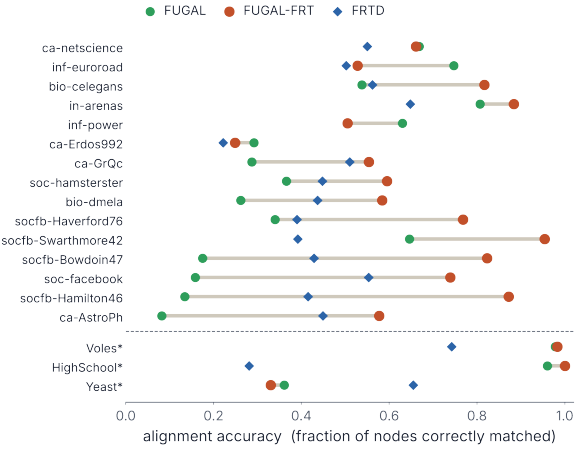
1 A. Bommakanti, H. R. Vonteri, K. Skitsas, S. Ranu, D. Mottin and P. Karras, *FUGAL: Feature-fortified unrestricted graph alignment*, NeurIPS 2024.

Results

University of Oxford
Mathematical Institute

Benchmark[1]

1. Copy a network G_1 to make G_2.
2. Delete 5% of G_2's edges at random.
3. Shuffle G_2's labels; this permutation is the ground-truth mapping.



1 K. Skitsas, K. Orlowski, J. Hermanns, D. Mottin and P. Karras, *Comprehensive evaluation of algorithms for unrestricted graph alignment*, EDBT 2023, pp. 260–272.

Network randomization

$$P_G(G') = \frac{e^{-\beta\, d(G,G')}}{Z}$$



$\beta = 0$

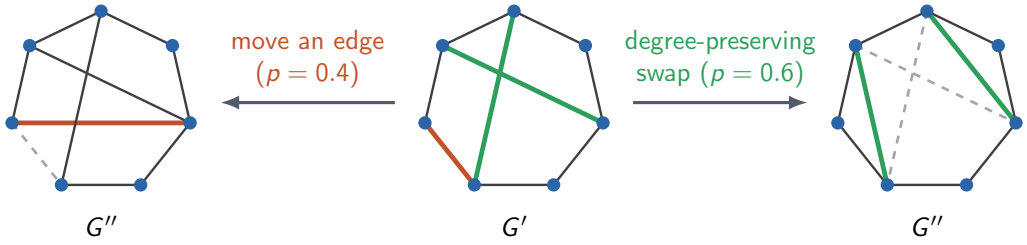
uniform over graphs with m edges: $G(n, m)$

$\beta \to \infty$

FRTD-matching graphs

Sampling the ensemble

Mathematical Institute

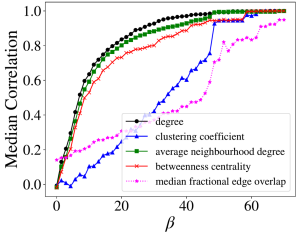


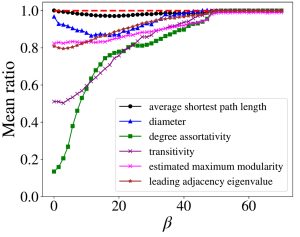
$$p_{\text{acc}} = \min\left(1,\ e^{-\beta\,\Delta d}\right)$$

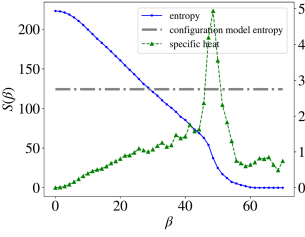
Results

Mathematical Institute

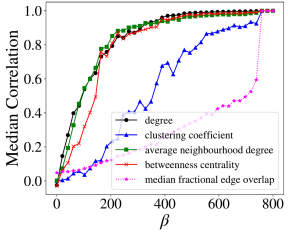
Karate club



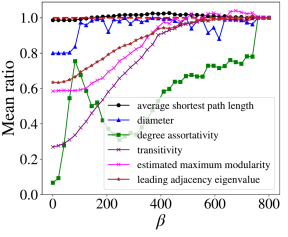




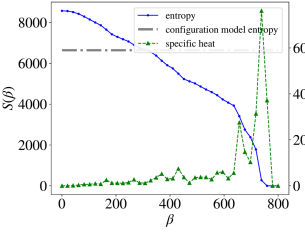
C. elegans



(a)



(b)



(c)

Conclusions

Mathematical Institute

- We have argued that the first return time distribution of a random walk captures meaningful structural information and forms a good graph embedding which is straightforwardly extended to weighted/directed networks.
- FRTD equivalence is stronger than cospectrality and corresponds instead to rotation by an orthogonal matrix.
- Demonstrated that the FRTD can be used to
  - define (and hence discover) structural roles;
  - identify nodes across corrupted versions of the same network;
  - define a distance between graphs, and generate random graphs still 'close' to the original.
- This provides further rationale for the existing uses of random walk statistics in machine learning, beyond the argument that 'they appear to work in practice'.

Open problems

- In the flavour of Haemers' conjecture, we conjecture that "almost all" graphs are determined by their FRTDs
- How do we efficiently generate graphs with prescribed FRTDs? the MCMC works in principle but is slow for large networks.
- *Scaling*: Exact FRTDs cost $O(Kn^2)$ or worse; Monte Carlo or message passing[1] give decentralized estimates but at the cost of accuracy.
- Other FRTD inspired embeddings of random walk statistics with stronger guarantees?

1 Hormann, Lambiotte & Cantwell, Phys. Rev. E (2025).

Thank you

Embedding networks with the random walk first return time distribution

Vedanta Thapar, Renaud Lambiotte, George T. Cantwell

Code & data
github.com/Vedanta-T/RandomWalk_FRTD_embeddings

Funding acknowledgement: Rhodes Trust

vedanta.thapar@maths.ox.ac.uk

Scan for the paper